Section D: Smart Contract Algorithm

```
BEGIN ALGORITHM: PCTF_Contact_Tracing
=====================================================================
// STEP 1: DEFINE THE INFECTIOUS WINDOW
SET infectious_period = 14 days
SET infectious_start  = DiagnosisDate - infectious_period
SET exposed_contacts  = EMPTY_LIST []

// STEP 2: SCAN CONTACT HISTORY FOR MATCHES
FOR EACH contact_entry IN ContactHistory[PatientKey] DO:

    IF (contact_entry.timestamp >= infectious_start) AND
       (contact_entry.timestamp <= DiagnosisDate) THEN

        IF verify_ipfs_record(contact_entry.CID) == VALID THEN

            SET target_key = contact_entry.contact_key
            ADD target_key TO exposed_contacts
            EXECUTE send_role_alert(target_key, PatientKey)

        END IF
    END IF
END FOR

// STEP 3: CREATE IMMUTABLE ON-CHAIN PROOF
SET diagnosis_signature = SHA256(PatientKey + DiagnosisDate)
EXECUTE record_on_chain(diagnosis_signature, "STATUS: DIAGNOSED")

// STEP 4: OUTPUT FINAL LIST
RETURN exposed_contacts
=====================================================================
END ALGORITHM
```